

Residential floor plans: Multi-conditional automatic generation using diffusion models

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ABSTRACT

Automatically generated residential floor plans using Artificial Intelligence that lower the skill barriers in and facilitate non-professional residential design, have become a significant topic. However, in previous studies, the limitations of RFP generative models have exhibited low controllability in outputs and limited flexibility in input conditions. In this study, a multi-conditional, two-stage generative model, FloorplanDiffusion, was developed to address these shortcomings. Based on Denoising Diffusion Probabilistic Models, a new model structure was established, allowing human designers to intervene for enhanced controllability. Furthermore, we implemented a multi-condition model input with structured information using images, thus significantly enhancing the model's input flexibility. Finally, through experiments we demonstrated that our model flexibly generates high-quality, diverse, and controllable results. A Turing test indicated that our model has the capacity of human experts.

1. Introduction

Residential plans represent the bulk of projects in the architectural, engineering, and construction (AEC) industries and auto-generating them using Artificial Intelligence (AI) has attracted increasing attention [1]. First, automatically generated residential floor plans (RFPs) can significantly improve the efficiency and accuracy of architectural design. Particularly in 3D CAD, during the conceptual phase, designers repeatedly draft, reflect, and modify, which is highly time-consuming [2]. Through auto-generation, designers can reduce the repetitive tasks, trial-and-error, and time costs in completing the design [3]. Second, auto-generated RFPs would alleviate the workload by eliminating tedious tasks such as layout, equipment configuration, and size adjustments, thus encouraging a stronger focus on creativity and quality. In addition, auto-generated RFPs are assured of design consistency and standardization.

Automation technology is already important in architectural design and throughout the AEC industries for improving design efficiency and quality, reducing design errors and operating costs, and promoting the sustainable development of residential design. The further promotion of research and development on architectural design automation and RFP auto-generation is paramount for the AEC industries.

Currently, most residential design relies on experienced designers who are required to make many revisions. The motivation for this study was to lower barriers in residential design to allow non-professionals to achieve reasonable designs while the professionals can make successive modifications to refine the RFPs. For this, the task has to have the following two functions, first, flexibility in the input conditions, because laymen cannot generate RFPs with the abstract inputs of professional designers, and their need to customize the requirements for their own residences, e.g., the size and configuration of the rooms as well as the edges of the residence's silhouettes. Second, the generated results must be stable, diverse, and controllable. Stability ensures that the quality of the output is excellent and not blurred, and that the generated results meet the user's customization requirements. Diverse allows numerous floor plans (FPs) to be generated for the user to choose from. Controllability means that the user can change all aspects of the generated FPs. If the generated result does not meet the user's requirements, the user can repeatedly and controllably modify the functional blocks and the contour edges of the design until it meets all requirements.

Although GAN and GNN models have generated RFPs, two aspects limit their use. First, the flexibility of input conditions must be improved. Second, the stability, diversity, and controllability of output FPs must be enhanced. In particular, GAN models are often difficult to

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train and encounter convergence issues, producing unstable results. In addition, these models have relatively low semantic information-capturing abilities. Conversely, GNN is convenient for professionals when designing RFPs, but the single input of graphs is not user-friendly for inexperienced non-professionals, and cannot ensure controllability when iterative modifications are made (e.g., the living room's shape remains unchanged to modify the size of the bedrooms). Furthermore, most current models adopt an end-to-end generation approach, which significantly affects controllability.

Through this analysis, we identified a lack of research on models for generating RFPs with high flexibility and controllability characteristics. To address this gap, this study investigated the potential of diffusion models as a BM for Generating RFPs to improve model flexibility and controllability. We analyzed and experimentally investigated their feasibility and methods regarding stability, diversity, flexibility, and controllability.

To our knowledge, this study is the first to adopt a diffusion model that addresses input flexibility and controllability while ensuring the stability and diversity of the generated results. The main contributions of this study are:

- The proposed FloorplanDiffusion model adapts to multi-condition input information, which significantly improves input flexibility whilst generating stable high-quality RFPs.
- Two-stage preprocessing is introduced, dividing the end-to-end process into generating the room-layout and wall, window, and door generation stages. Combined with preprocessing two-dimensional Gaussian gradient probability location information, the model captures additional RFP features while enhancing the controllability of FP generation.
- In extensive validation experiments, FloorplanDiffusion produced an FID score of 8.36 on the RPLAN dataset, a 74.2% improvement compared to baseline generative models. The Turing test reached an accuracy of 54.4%, indicating that the proposed model reached expert levels in generating RFPs.

2. Related works

2.1. Big models

2.1.1. Rise of deep learning (DL) and big models (BM)

In recent years, the rapid development of DL and BM [4] has gained significant attention. Their excellent recognition and classification capabilities have seen DL algorithms widely applied in machine vision, natural language processing (NLP), and sequential prediction among others. Convolutional Neural Networks (CNNs) [5] have been extensively used for image recognition and classification tasks, Recurrent Neural Networks (RNNs) [6] have been applied to speech and NLP, and Deep Reinforcement Learning (DRL) [7] has been employed for autonomous decision-making and control in intelligent systems. BMs utilize a substantial number of parameters for training, in addition to DL, where the interlayer weights of neurons can reach billions. BM technology has also been widely applied owing to its high accuracy and reliability. For example, BM such as Bidirectional Encoder Representations from Transformers (BERT) [8] and Generative Pre-trained Transformer (GPT) [9] have been extensively used in NLP. OpenAI released ChatGPT [10], a large-scale language model (LLM) based on the transformer architecture that is capable of generating human-like text based on context and past conversations [11]. AlphaGo [12] demonstrated superior Go-playing abilities and outperformed the best human players. The Mid-Journey BM [13] has shown exceptional results in image generation. Ghost-Mix, a BM, achieved an average rating of 4.99 out of 5, based on 370 image generation evaluations [14]. The DL and BM technologies are expected to yield ever more significant achievements in many domains thus contributing to solving various real-world problems.

2.1.2. Opportunities for BM applications in generating RFPs

AI and DL have gradually infiltrated architectural design. Traditionally, designs have been produced by professional architects, who require copious time and are susceptible to the limitations imposed by experience and skill. However, DL technology has provided new directions and opportunities for automation and intelligence in architectural design.

Among these, GAN models [15] are innovative architectures that have recently achieved significant breakthroughs in DL. Currently, GAN model applications in automatic architectural design have become a key research topic [16–18]. Simultaneously, GNN models have attracted widespread attention and research as a new architecture in the RFP design field. GNNs can generate RFPs with reasonable functional relationships by processing and learning the graph structure information of RFP functional relationships [19] (details in Section 2).

Although GAN and GNN models have achieved results in generating RFPs, improvements are desirable regarding their stability, diversity, flexibility, and controllability. Contrastingly, GNN models offer greater convenience to professional designers, but the work in this paper is aimed at non-professionals, whose need for input flexibility to customize the output (size and shape of room blocks and contour edges, etc.) contradicts the task with graph-based GNN models. In addition, the current end-to-end model structure of GNN models lacks controllability for generating RFPs. BMs have many more parameters and may have significant potential for stability, variety, flexibility, and controllability in generating RFPs. Therefore, the introduction of BM technology into the task of automatically generating RFPs requires further investigation. This research specifically aims to address that gap.

2.2. Generative models

GANs [15], proposed by Goodfellow et al., have been extensively applied to image generation. For example, the classic Pix2Pix model [20] and conditional GAN (cGAN) [20] are image-to-image transformation models. Subsequently, Zhu et al. proposed CycleGAN [21], which achieved unpaired image translation using adversarial training. GAN has been applied to a shear wall design for significant acceleration and quality improvement [22]. Additionally, it was used for automatically generating [23] visually realistic and semantically meaningful architectural layouts. GNN models, a class of generative models for handling graph-structured data, have recently undergone rapid development. The key to success for these models lies in node representation learning and the propagation of graph structure information. For example, the GraphRNN model proposed by Lin et al. [24] is a deep autoregressive model that generates realistic graph data. Other models combine GNN structures, e.g., the GraphGAN model includes a GAN [25], the GraphVAE model incorporates variational autoencoders (VAEs) [26], and the GraphAF model is based on flows [27]. As GNNs evolve, their derivative models have established extensive architectural applications. Li et al. proposed an automated process using a GNN and exploratory genetic algorithms [28] for reinforced steel design optimization. Furthermore, with the development of Transformer [11] and diffusion models [29], generative models have found new opportunities for development. In NLP, the latest advancement in generative models is ChatGPT, which generates coherent natural text. Transformer architecture has also performed well in architectural 3D model-aided design [30]. The Midjourney model, [13], is a novel generative model with excellent performance. Recently, diffusion models have shown superior performance in generating stable high-quality images. During training, this model fits the input data with unknown distributions by characterizing the underlying data distribution, thereby enabling high-quality and stable image generation. A recent diffusion model based on a transformer structure offers new concepts and methods to achieve realistic image generation [31].

In summary, the introduction of various model frameworks has provided powerful tools and theoretical support for generating images,

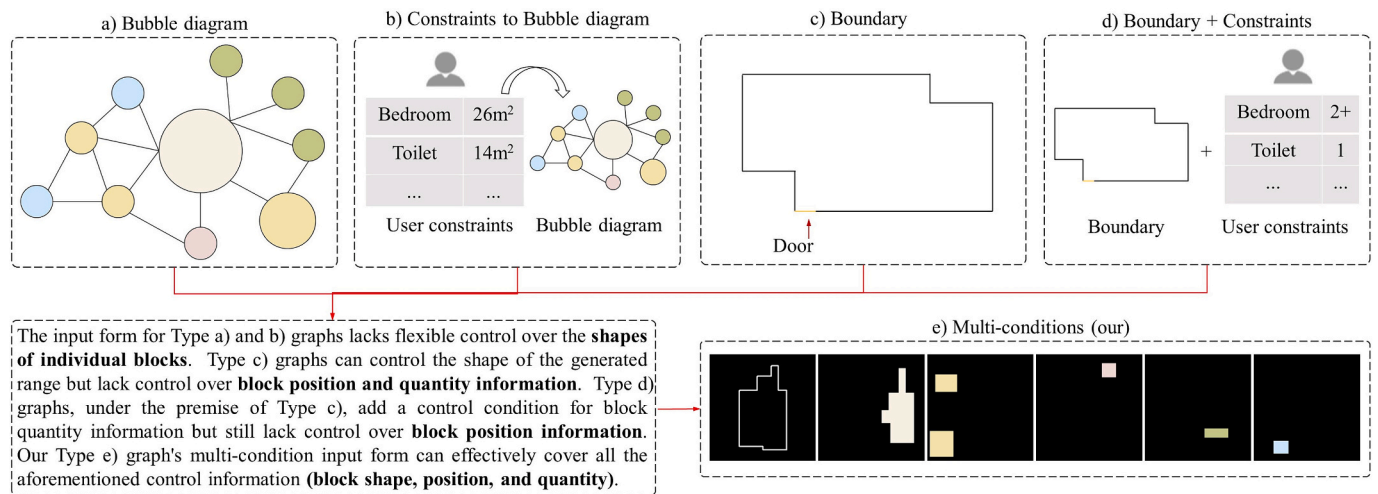


Fig. 1. Input forms for generating RFPs. a) and b) Graphs represent input forms based on bubble diagrams and boundaries. c) Graphs represent the input form adding room number constraints to boundaries. d) Graphs represent the input form for generating bubble diagrams by constraining room areas. e) Graphs represent the proposed multi-condition input form.

graph data, and text.

2.3. Generative model application in RFP auto-generation

In previous research, Wu et al. (2019) proposed a data-driven approach to automatically generate interior FPs that adhere to given boundaries [32]. Nauata et al. (2020) introduced a relation-based GAN to generate residential building layouts that satisfy graph constraints [33]. Hu et al. (2020) proposed an image-based generative model that learns the spatial layouts and relationships of objects in images to generate FPs with specific styles [19]. Para et al. (2021) proposed a novel approach using constraint graphs for layout modeling [34], resulting in higher-quality outputs. Nauata et al. (2021) developed a GAN for layout optimization that automatically generates FPs [35] via the iterative optimization of previously generated layouts. Huang et al. (2021) utilized GAN and Pix2PixHD models to recognize and generate architectural drawings [36]. Additionally, other methods exist, such as style transfer [37] and the use of denoising diffusion probabilistic models (DDPMs) [38] for generating vector FPs, which have achieved significant progress in generating RFPs. Simultaneously, significant advances have recently emerged in other aspects of RFP design, such as wall segmentation [39], FP reconstruction, [40] and the handling of non-cuboidal room layouts [41], which have shown promising results.

However, existing methods for generating RFPs are limited regarding the input of each model. Current models predominantly rely on bubble diagrams, boundaries, and the addition of explicit constraint conditions (Fig. 1). The input for bubble diagrams is mainly in the form of a graph, which is convenient for professional designers, but may be difficult and abstract for non-professionals to understand because the input of information such as shapes is lacking and the difficulty of controllably while editing the generated RFPs. Even with the addition of textual requirement constraints, accurately constraining the shapes of residential properties remains challenging. Consequently, these models exhibit limitations regarding input flexibility.

3. FloorplanDiffusion model and methodology

3.1. Workflow and framework

This study proposed a multi-condition, two-stage model called FloorplanDiffusion, based on a diffusion model that achieves high controllability and flexibility in generating RFPs. This provides users with flexible multi-condition input possibilities. This significantly

reduces input constraints and better meets users' diverse and personalized requirements. This model, which also enables flexible control over the output RFP comprises three parts: data input preprocessing, two-stage model training, and model evaluation and analysis, Fig. 2 illustrates the research framework.

During data input preprocessing, the input data are enhanced to ensure increased flexibility of the input conditions, thereby allowing diversified RFPs to be generated in the subsequent model. Features are extracted from the input images and image data is augmented to capture hidden features better during model training. Throughout, the model controls the output images by capturing the semantic image information when encountering different input conditions. Consequently, the developed model is more flexible and controllable when generating RFP images. For the two-stage model training, RFP image generation is divided into two manually modifiable stages: room layout generation and wall, window & door generation.

3.2. Data preprocessing

Data preprocessing ensures that the input conditions are more flexible and provides certain feature information for generating controllable and diversified RFPs as shown in Fig. 3.

Input data preprocessing is divided into five steps. Step 1: The RFP map is downsampled. Steps 2 and 3: Two-stage preparation and semantic separation are performed on the RFP. Step 4: Embed the gradient position information into the processed image in Step 4 and finally, Step 5: Data augmentation preprocessing. Using these five steps, the input data are preprocessed thus enabling the model to capture features and flexibly generate diverse RFPs.

3.2.1. Resampling

To reduce the model training cost, a pixel-region-based image resampling strategy is adopted. Specifically, the original image resolution is downsampled from 256 to 64 pixels. By reducing the resolution, the number of pixels that must be processed can be reduced, thereby lowering the computational load of the model while preserving essential information.

3.2.2. Two-stage processing

Typically, end-to-end trained AI models (such as GAN and GNN) directly generate the final design. However, constrains human control over the generation process. To address this challenge, two-stage processing is employed. In the first stage, the input consists of multi-

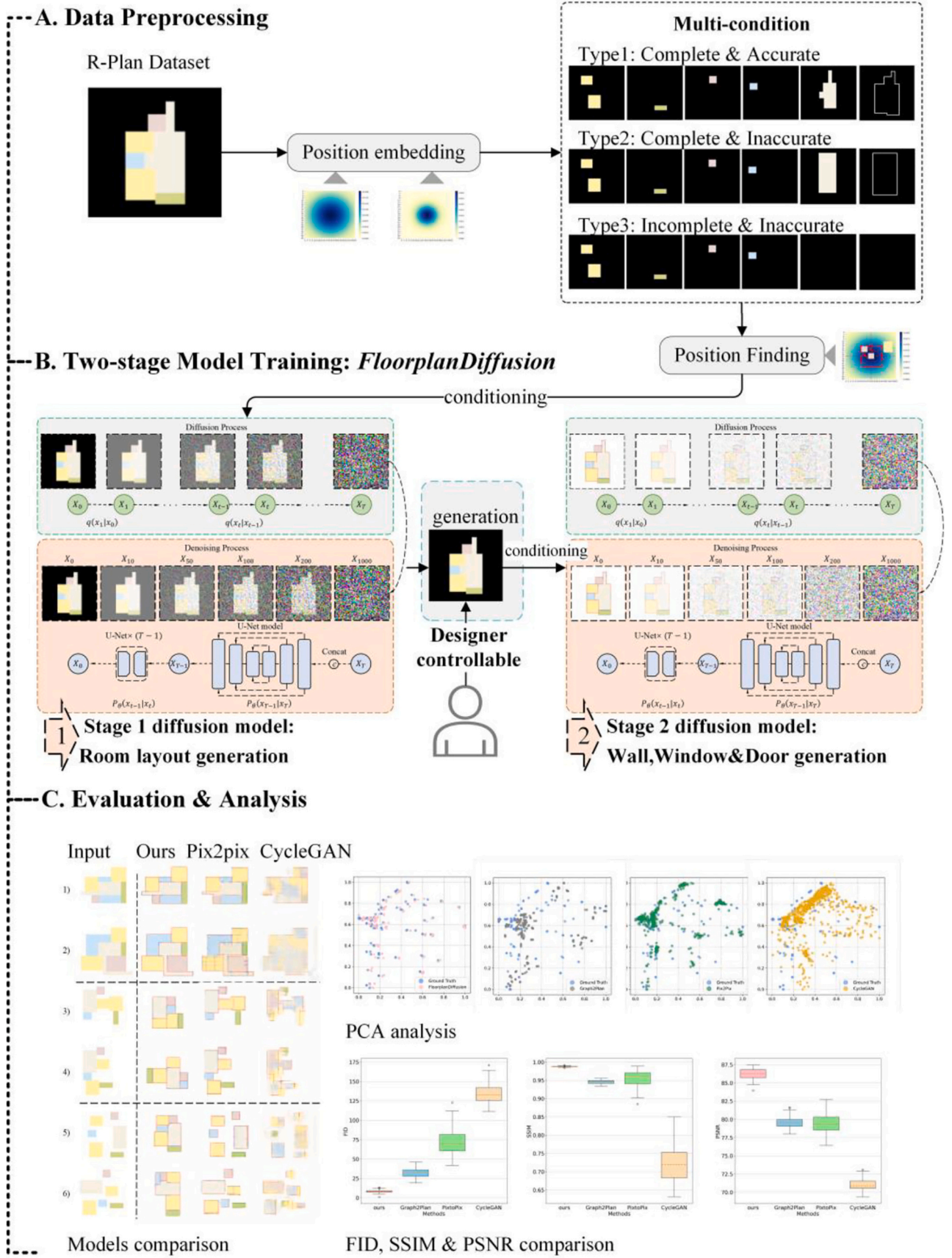


Fig. 2. The FloorplanDiffusion model's framework.

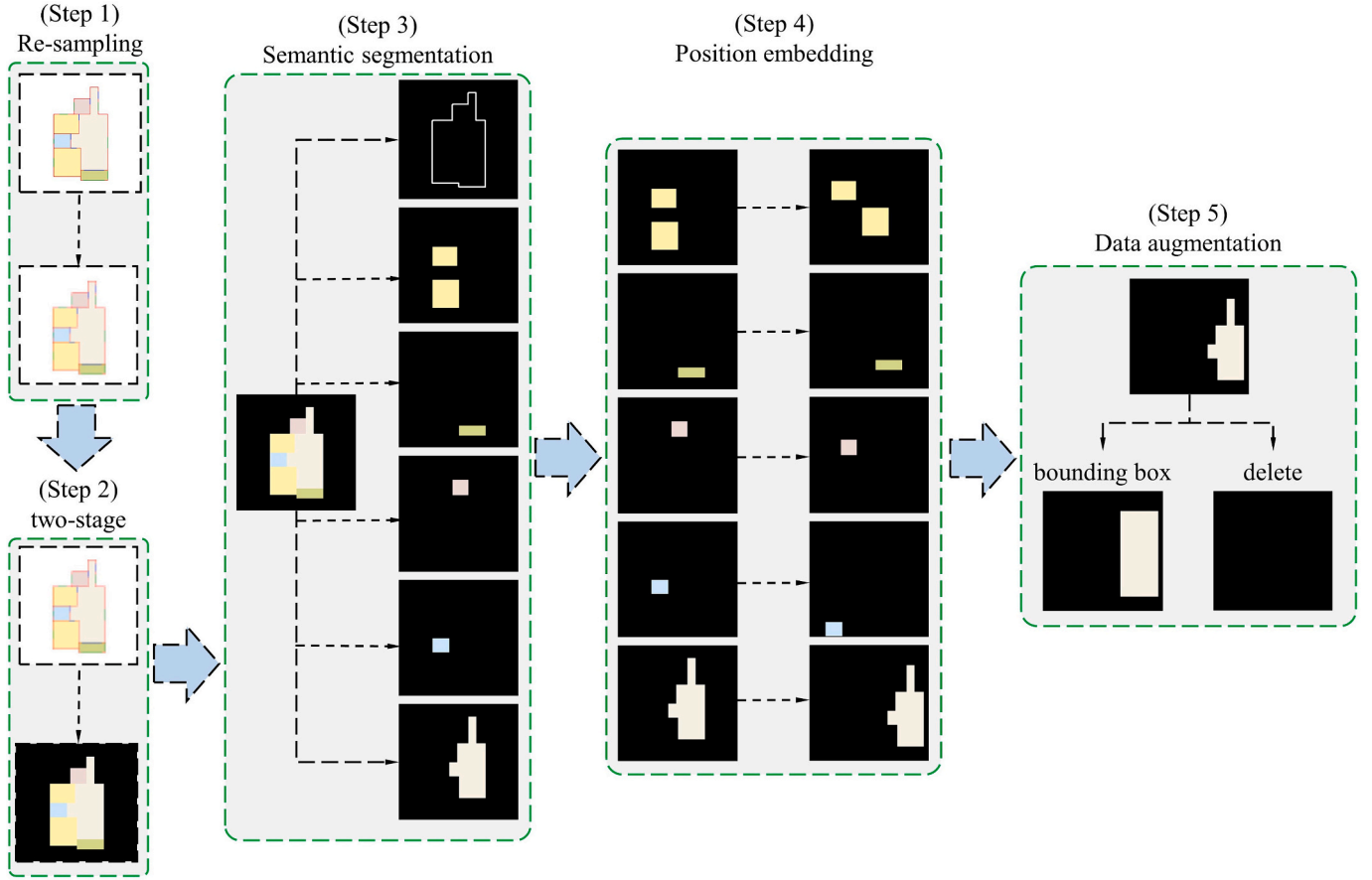


Fig. 3. Data preprocessing flowchart.

condition input blocks that have undergone semantic separation and include positional information. The output of the first stage which comprises the input for the second stage is FP images with reasonable positional information. The output of the second stage is a complete RFP. Using two-stage processing, we can modify the generated FPs at each stage, resulting in stable rational designs with enhanced model controllability.

3.2.3. RFP semantic separation

In designing the RFP, we adopted a semantic separation method to divide the complete FP image into different types of semantic information, decomposing it into five input images each of which represents a different semantic category. Specifically, the different colored blocks correspond to different room types: gray represents living rooms, yellow represents bedrooms, green represents balconies, pink represents kitchens, and blue represents bathrooms, as shown in Fig. 3. This semantic separation approach sets the foundation for capturing semantic image features using generative models, thus enabling multi-condition model inputs (living rooms, etc.) with improved flexibility.

3.2.4. Positional information embedding

To introduce structured positional information and control relative positional changes during the generation process, we designed a two-dimensional Gaussian gradient probability-based positional information preprocessing mechanism. Specifically, we used the center point of each block as the Gaussian mean and applied random position movements based on a two-dimensional Gaussian distribution.

Therefore, we must first calculate the center points of the blocks. Specifically, the calculation formulae for the x- and y-axis center points of the blocks are as follows:

$$(x_{central}, y_{central}) = \left(\frac{2x_{pic} + w_{pic}}{2}, \frac{2y_{pic} + h_{pic}}{2} \right)$$

Among these, $x_{central}, y_{central}$ represent the center points of the x, y axes of the block. x, y represent the x_{pic}, y_{pic} coordinates of the top-left corner bounding box of the block. w_{pic}, h_{pic} represent the block-bounding box width and height.

After obtaining the block center point, we used a two-dimensional Gaussian probability distribution considering the block center as the mean to move the block's position. The reason for this is that the user includes structural data information, i.e., information such as the location and orientation of the block, in the given input block image. The model should search for the best location in the vicinity of the block based on the above information given by the user. Therefore, the mean of the 2D Gaussian distribution is the center point of the block and the larger the variance, the larger the search range. Specifically, the following probability distribution formula was used:

$$f_{\mathbf{x}}(x_1, \dots, x_k) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)}, (k=2)$$

$$\mu = \begin{pmatrix} x_{central} \\ y_{central} \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}$$

In the aforementioned equation, σ_x, σ_y are the sets of hyper-parameters that represent the standard deviations of the two-dimensional Gaussian probability distribution. A larger standard deviation implies a wider probability distribution range of positional movement and a weaker constraint on the relative positional information. Conversely, a smaller standard deviation indicates a narrower

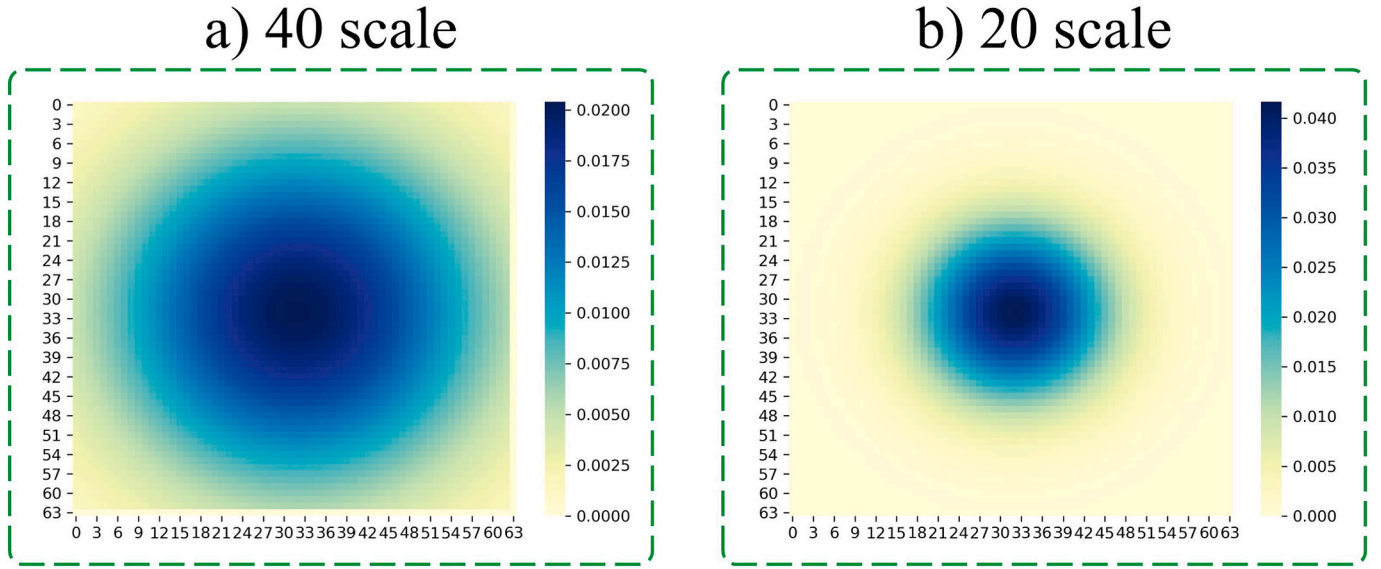


Fig. 4. Two-dimensional Gaussian probability illustration. The image a) had a standard deviation of 40, whereas that b) had a standard deviation of 20.

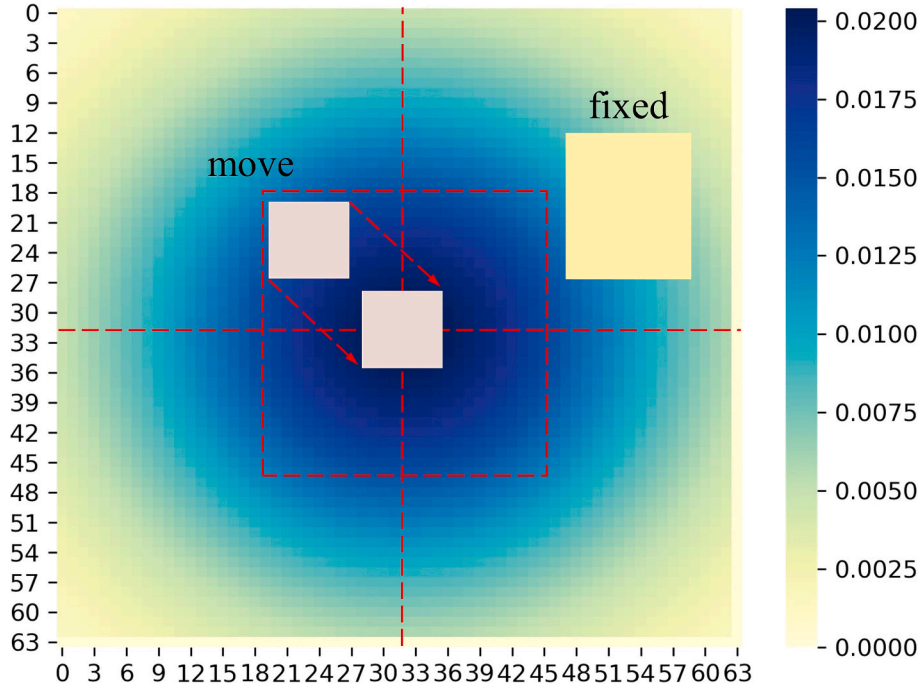


Fig. 5. Steps for identifying spatial relationships during model generation. The pink block in the middle and the yellow block in the upper right-hand corner respectively represent the kitchen and bedroom. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

probability distribution range for positional movement and a stronger constraint on the relative positional information. ρ represents the Pearson product-moment correlation coefficients (PCCs), which indicate the degree of linear correlation between variables x , y . This hyperparameter was set to zero.

To capture the relative positional features of different gradients, we controlled the strength of the position information constraint by setting the gradients of the σ_x, σ_y hyperparameters. In this study, we considered examples with standard deviations of 20 and 40, as shown in Fig. 4.

3.2.5. Data augmentation

To enhance model generalizability and promote diversity in the

generated outputs, we employed a series of data augmentation techniques. Bounding box and deletion processing were performed on semantic blocks to adapt them to different types of inputs, namely, Type 2 (complete & inaccurate) and Type 3 (incomplete & inaccurate). Thus, we also reduced the input constraints to support diversity in the output. These data augmentation methods provide additional input samples to better understand and respond to various scenarios, thereby underpinning the model's generalizability and diversity.

3.3. Two-stage model training

Based on diffusion, this section proposes a two-stage model, which

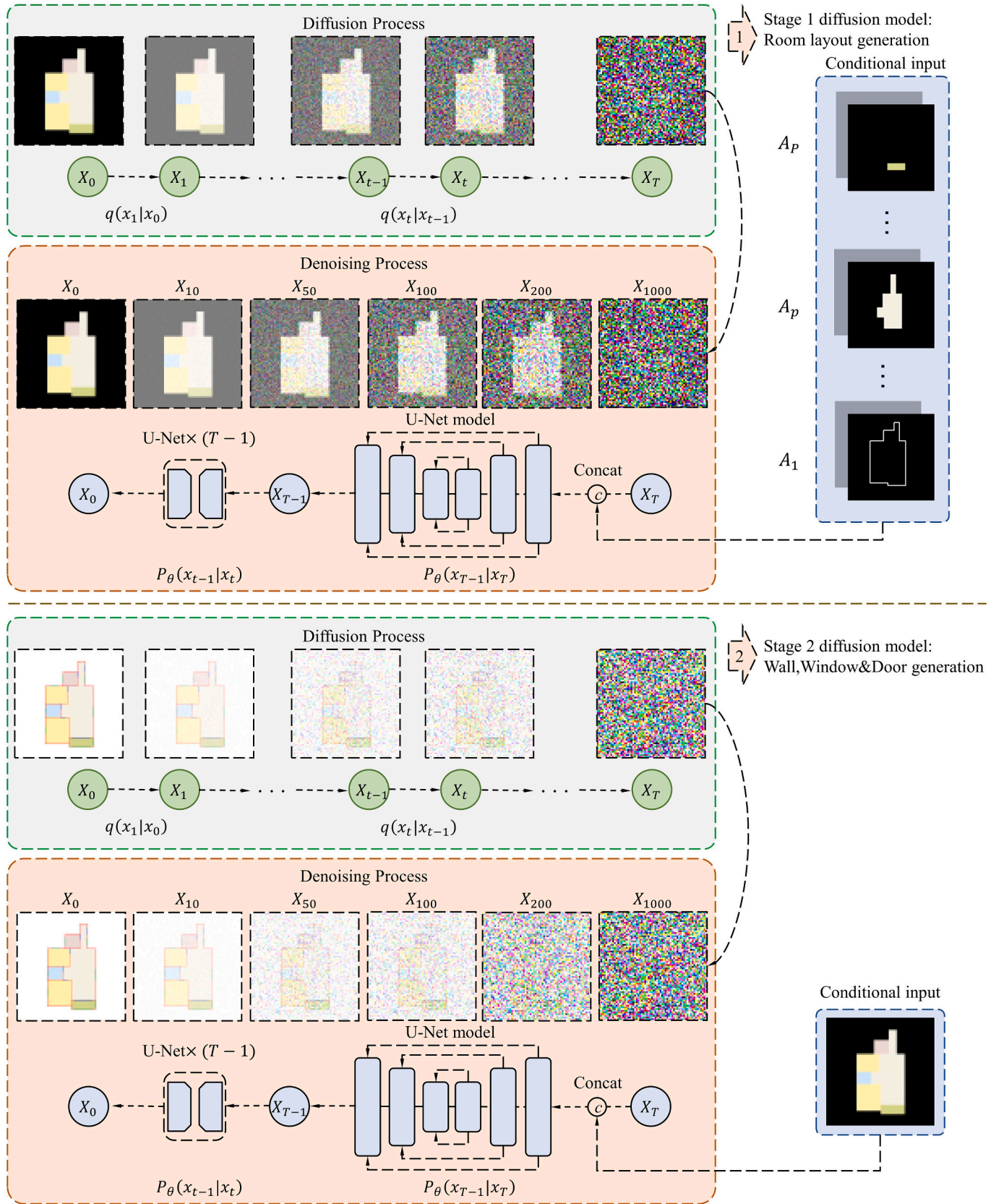


Fig. 6. Structure of the FloorplanDiffusion model's generation stage. The generation stage was divided into two steps: Stage 1 - room layout generation, and Stage 2 - wall, window, and door generation. Each stage consists of a diffusion process and a reverse diffusion process, a U-Net network was utilized to iteratively fit the noise in the diffusion process during reverse diffusion.

decomposes the image processing procedure of traditional end-to-end generation into two-stage generation. The first stage generates an FP layout, and the second generates a complete RFP (with walls, windows, and doors).

3.3.1. Two-stage model

3.3.1.1. Identifying positional relationships. The primary goal of this stage is to analyze the positional information block features. We determined whether to limit the block's positional information by utilizing a rectangular region surrounding the image center (Fig. 5). If the center of the block falls within this region, it is relocated to the image center. For blocks positioned outside the rectangular region, we assume that the user has prior knowledge of their positions; therefore, their positional information is not modified.

3.3.1.2. Two-stage generation. In the room-layout generation stage, we used multi-conditioned images (a set of images that each included the information of a type of rooms such as living rooms and bedrooms) as the model input and generated floor plans with reasonable relative positioning relationships using the model. In the wall, window, and door generation stage, we used the resulting room layout as the model input and generated complete RFPs using this model. The modeling process is illustrated in Fig. 6.

The diffusion model was adopted for image generation. The working principle of this model is that Gaussian noise is continuously added to destroy the training data, transforming it into a pure noise distribution $N(0, I)$, then, denoising fits the noise distribution to restore the data. Diffusion models can be divided into the forward diffusion and the backward inverse diffusion types.

- 1) Forward diffusion: This refers to adding noise to an image. Although this step cannot generate an image, it is crucial to understand the diffusion models and construct training samples for generating RFPs.

Given a real image sample $x_0 \sim q(x)$, forward diffusion accumulates Gaussian noise up to T times, resulting in $x_1, x_2, \dots, x_T$, as shown in the q process in Fig. 6. The size of each step is controlled by the hyperparameter $\{\beta_t \in (0, 1)\}_{t=1}^T$, which represents a series of Gaussian distribution variances. Because at each moment t the forward process only depends on $t-1$, it can also be regarded as a Markov process:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I})$$

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1})$$

In this process, as t increases, x_t approaches being pure noise. When $T \rightarrow \infty$, x_T is entirely Gaussian noise (with a choice of mean coefficient $\sqrt{1-\beta_t}$).

This approach requires random sampling from a Gaussian distribution; however, it cannot backpropagate the gradient. Simultaneously, diffusion requires multiple samplings from Gaussian noise to obtain x_t , thus, we must use a reparameterization technique to ensure that it is differentiable.

Typically, the approach for (e) is to guide randomness through an independent random variable. That is, if we wish to sample z from a Gaussian distribution $z \sim \mathcal{N}(z; \mu_\theta, \sigma_\theta^2 \mathbf{I})$, it can be expressed as follows:

$$z = \mu_\theta + \sigma_\theta \odot \epsilon, \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

The value of z in the aforementioned equation continues to remain stochastic and follows a Gaussian distribution with a mean of μ_θ and a variance of σ_θ^2 . The values of μ_θ and σ_θ^2 are inferred by the neural network parameters θ . The entire sampling process continues to be differentiable with respect to gradients, where the randomness is transferred but not parameterized for training. Therefore employing the aforementioned

reparameterization technique circumvents the issue of gradient non-backpropagation.

Simultaneously, the forward process has a favorable property, where x_t can be directly obtained through x_0 and β . Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t =$

$\prod_{i=1}^t \alpha_i$ be defined; with the expansion of x_t , we obtain [29]:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

Therefore, at any given moment x_t satisfies:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I})$$

- 2) Reverse diffusion: The previous section discussed the forward process, also known as the noisy process. In this section, the reverse process refers to the diffusion denoising inference process.

If we reverse the noisy process and sample from $q(x_{t-1}|x_t)$, we can restore the distribution $x_0 \sim q(x)$ of the original RFP from the Gaussian noise $x_T \sim \mathcal{N}(0, I)$. It has been demonstrated that if $q(x_t|x_{t-1})$ satisfies a Gaussian distribution and β_t is small enough, $q(x_{t-1}|x_t)$ remains a Gaussian distribution [42]. However, we cannot directly infer $q(x_{t-1}|x_t)$, therefore, we use a DL model (with parameters θ and utilize the U-Net + attention structure) to predict such a reverse distribution p_θ (similar to VAE).

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t)$$

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

The variance of the conditional probability $p_\theta(x_{t-1}|x_t)$ is related to β_t , hence, a DL network does not need to estimate $\Sigma_\theta(x_t, t)$ and must only estimate the mean.

In summary, the forward and backward diffusion processes are both based on the Markov framework. The only difference is that in forward diffusion, the mean and variance of each conditional Gaussian probability distribution are already determined (dependent on β_t and x_0), whereas for the backward diffusion process, the mean and variance are learned through neural networks (with the variance related to β_t , such that it is only necessary to learn the mean).

Next, we derived the backward diffusion conditional probability. We cannot obtain the probability distribution of $q(x_{t-1}|x_t)$ in the backward diffusion process; however, we can derive the following equation based on Bayes' formula [29].

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

which:

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\begin{aligned} \mu_t(x_t, x_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0 \\ &= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_t \right) \end{aligned}$$

According to this equation, under the given condition of x_0 , the mean of the posterior conditional Gaussian distribution is only related to the hyperparameters, x_t, ϵ_t while the variance is only related to the hyperparameters.

The output image was obtained using the inverse diffusion process. During the training stage, the distribution of the input image of the diffusion process and the output image of the inverse diffusion process should be as consistent as possible. During the inference stage, the output of the inverse diffusion process represents the inferred output

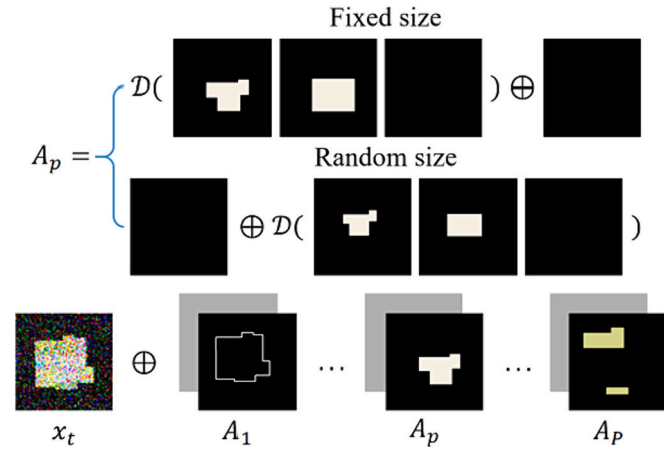


Fig. 7. Conditional dropout mechanism. The model has six conditions (A_p): exterior profile, living room, bedroom, kitchen, bathroom, and balcony. Each condition has three forms of input: zero values, raw values, and bounding box values.

Table 1

Proposed FloorplanDiffusion model test results compared to the baseline models on the RPLAN dataset evaluated using specific metrics.

Method	FID	PSNR	SSIM
CycleGAN	134.79	70.98	0.72
Pix2pix	71.39	79.43	0.95
Graph2Plan	32.45	79.57	0.95
FloorplanDiffusion	8.36	86.25	0.99
Truth	0.00	+Inf	1.00

image.

Finally, to obtain the final image the output data is post-processed, which includes image denoising and color block normalization in the generated RFP.

3.3.1.3. Conditional dropout. To adapt the model to the requirements of different residential layout designs at a low cost, we drop some model input conditions (Fig. 7). That is, $\mathbf{x}_t' = \mathbf{x}_t \oplus \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{p \times D}$ denotes the input conditions after dropout and $\oplus$ indicates image splicing. $\mathbf{A}_p = \mathcal{Z}(\mathcal{D}(\alpha_i^p, \beta_i^p, \gamma_i^p)) \in \mathbb{R}^D$, where $\alpha_i^p, \beta_i^p, \gamma_i^p$ means that p-type room inputs are respectively zero values, original values, or bounding-box values (the room shape is transformed into a bounding box). We consider two cases for room size ($D = 2$): variable or non-variable. Function $\mathcal{D}(\cdot)$ randomly deletes multiple conditions, and only one condition is selected. In $\mathcal{Z}(\cdot)$, if the state set by $\mathcal{D}(\cdot)$ belongs to the room size variable, then $\mathbf{A}_p = \text{Zero} \oplus \mathcal{Z}(\alpha_i^p, \beta_i^p, \gamma_i^p)$; otherwise, $\mathbf{A}_p = \mathcal{Z}(\alpha_i^p, \beta_i^p, \gamma_i^p) \oplus \text{Zero}$, where Zero is a zero-valued image. Thus, the inverse diffusion process of the model is

$$p_\theta(x_{0:T}) := p_\theta \prod_{t=1}^T p_\theta(x_{t-1}|x_t)$$

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \sum_\theta(x_t, t)\right).$$

3.3.1.4. Output control. For our model generation process, the original end-to-end one-step generation of the RFP is decomposed into two stages. This allows us to modify the inputs of both stages of the generative model separately and ultimately control the generated RFP. There are two specific modification methods:

- 1) Modifying the first stage multi-condition input. By transforming one or more semantic blocks into other shapes or by completely removing certain semantic blocks, we can regenerate the RFP using the two model stages.
- 2) Modifying the input during the second stage. Here, the generated room layout is manually modified and used as an input for the second stage to generate the RFP again through the model.

Thus, we can further adjust and modify the RFPs manually. Using these two methods, we can control the generation process of RFPs more precisely, thus being able to meet more specific requirements and constraints.

3.3.2. Model training

3.3.2.1. Training loss function. The purpose of training a diffusion model is to determine the probability distribution of Markov chain transitions in inverse diffusion processes by estimating maximum likelihoods. Expressed differently, maximizing the logarithmic likelihood of the model's predicted distribution can be perceived as minimizing the negative logarithmic likelihood.

$$\mathcal{L} = \mathbb{E}_q[-\log p_\theta(x_0)]$$

This is similar to the VAE, therefore, we can use the variational lower bound (VLB) to optimize the negative log-likelihood as follows:

$$\begin{aligned} -\log p_\theta(x_0) &\leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0)) \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \end{aligned}$$

In this case, we let:

$$\mathcal{L}_{\text{VLB}} = \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \geq -\mathbb{E}_{q(\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0)$$

Through derivation, $\mathcal{L}_{\text{VLB}}$ is the sum of multiple KL divergences.

$$\begin{aligned} \mathcal{L}_{\text{VLB}} &= \mathbb{E}_q[L_T + L_t - L_0] \\ L_T &= D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T)) \\ L_t &= \sum_{i=2}^T D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ L_0 &= \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \end{aligned}$$

In this equation, the first term is the L_T term, the second is the L_t term, and the third is the L_0 term. Because the q forward process has no trainable parameters and $\mathbf{x}_T$ is Gaussian noise, the L_T term can be considered a negligible constant. Furthermore, after derivation, the second term L_t can be expressed as:

$$\begin{aligned} L_t &= \mathbb{E}_q[\|\mu_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2] \\ &= \mathbb{E}_{x_0, \epsilon} \left[\left\| \frac{1}{\sqrt{\alpha_t}} \left(x_t(x_0, \epsilon) - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon \right) - \mu_\theta(x_t(x_0, \epsilon), t) \right\|^2 \right] \epsilon \sim \mathcal{N}(0, 1) \end{aligned}$$

here $\mu_\theta(x_t, t)$ is directly predicted using neural networks. However, it is evident that $\mu_\theta(x_t, t)$ is related to noise ϵ . Therefore, we directly predicted noise ϵ using the neural network $\epsilon_\theta(x_t, t)$, and subsequently input the predicted noise into the defined expression to calculate the predicted mean. Therefore, the final representation of L_t is:

$$L_t = \mathbb{E}_{x_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, t \right) \right\|^2 \right], \epsilon \sim \mathcal{N}(0, 1)$$

Finally, L_0 can be regarded as a data reconstruction, and optimization is achieved using a negative log-likelihood. L_0 uses the estimated $\mathcal{N}(x_0; \mu_\theta(x_1, 1), \sum_\theta(x_1, 1))$ to construct a discrete decoder for computation.

3.3.2.2. Model parameter settings. To optimize the convergence speed



Fig. 8. Sample of the model's generation results.

and stability of the model, the initial learning rate is set to $2e-4$, and this learning rate is gradually reduced using a learning rate decay strategy. We set the batch size to 128 to fully utilize the computational resources and accelerate optimization.

3.3.2.3. Training and tuning. For tuning, we selected the best model parameters using cross-validation and an evaluation of the performance metrics on a validation set. We conducted several experiments and adjusted these parameters to determine the optimal combination. Through continuous experimentation and optimization, we successfully improved the performance of the model, thus producing improved results in generating RFPs.

3.3.2.4. Training platform. All experiments were conducted on an Nvidia A6000 48 G GPU for training and testing.

3.4. Evaluation and analysis

The model was analyzed by validating the metrics and serially evaluating the generated images while the Turing tests were conducted using expert ratings.

3.4.1. Evaluation metric

This section describes the evaluation methods used to validate the effectiveness of the proposed model. To evaluate the generative model comprehensively, the performance had to meet all criteria. Therefore, we chose various evaluation metrics, including the Fréchet Inception Distance (FID), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM).

3.4.1.1. FID metric. FID captures the differences in feature distribution between the generated and real images, thereby evaluating the quality of the generated model. Our goal was to minimize the FID value to indicate a smaller distribution difference between the generated and real

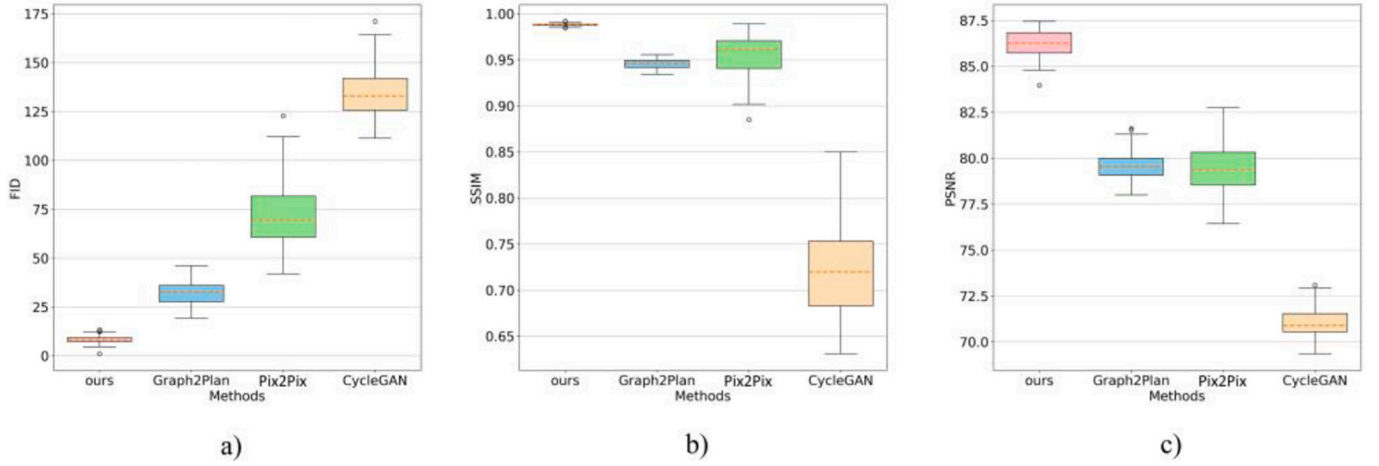


Fig. 9. Model stability analysis graph for the FID, PSNR, and SSIM metrics.

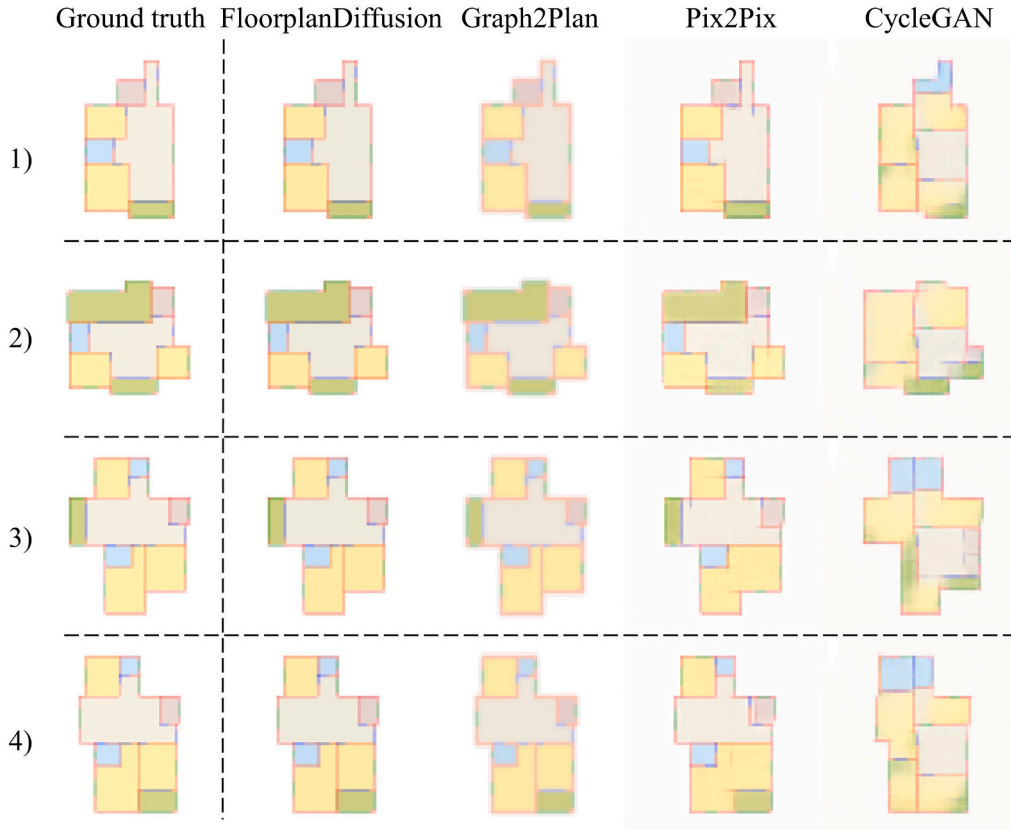


Fig. 10. Model generation comparison examples. The first column represents the ground truth, whereas the remaining four columns show the proposed model and other baseline models.

images, leading to better generation performance. The formula for calculating the FID metrics is:

$$FID(g, r) = \|\mu_g - \mu_r\|_2^2 + \text{Tr}(\Sigma_g + \Sigma_r - 2(\Sigma_g \Sigma_r)^{\frac{1}{2}})$$

where g and r represent the generated and real images, respectively, while μ_g and μ_r denote the means of their respective feature vectors. Σ_g and Σ_r represent the covariance matrices of the feature vectors and $\text{Tr}(\cdot)$ represents the trace of the matrix.

3.4.1.2. PSNR metric. The PSNR evaluates the fidelity of an image by

calculating the mean square error (MSE) between the original and reconstructed image. A higher PSNR value indicates a smaller difference between the reconstructed and original images, indicating a higher quality of reconstruction. The formula for calculating the PSNR is:

$$PSNR = 10 \cdot \lg \left(\frac{MaxValue^2}{MSE} \right)$$

In this formula, MSE represents the mean squared error between two images, and $MaxValue$ represents the maximum value the image pixels can take, e.g., $2^8 - 1 = 255$ for an 8-bit image.

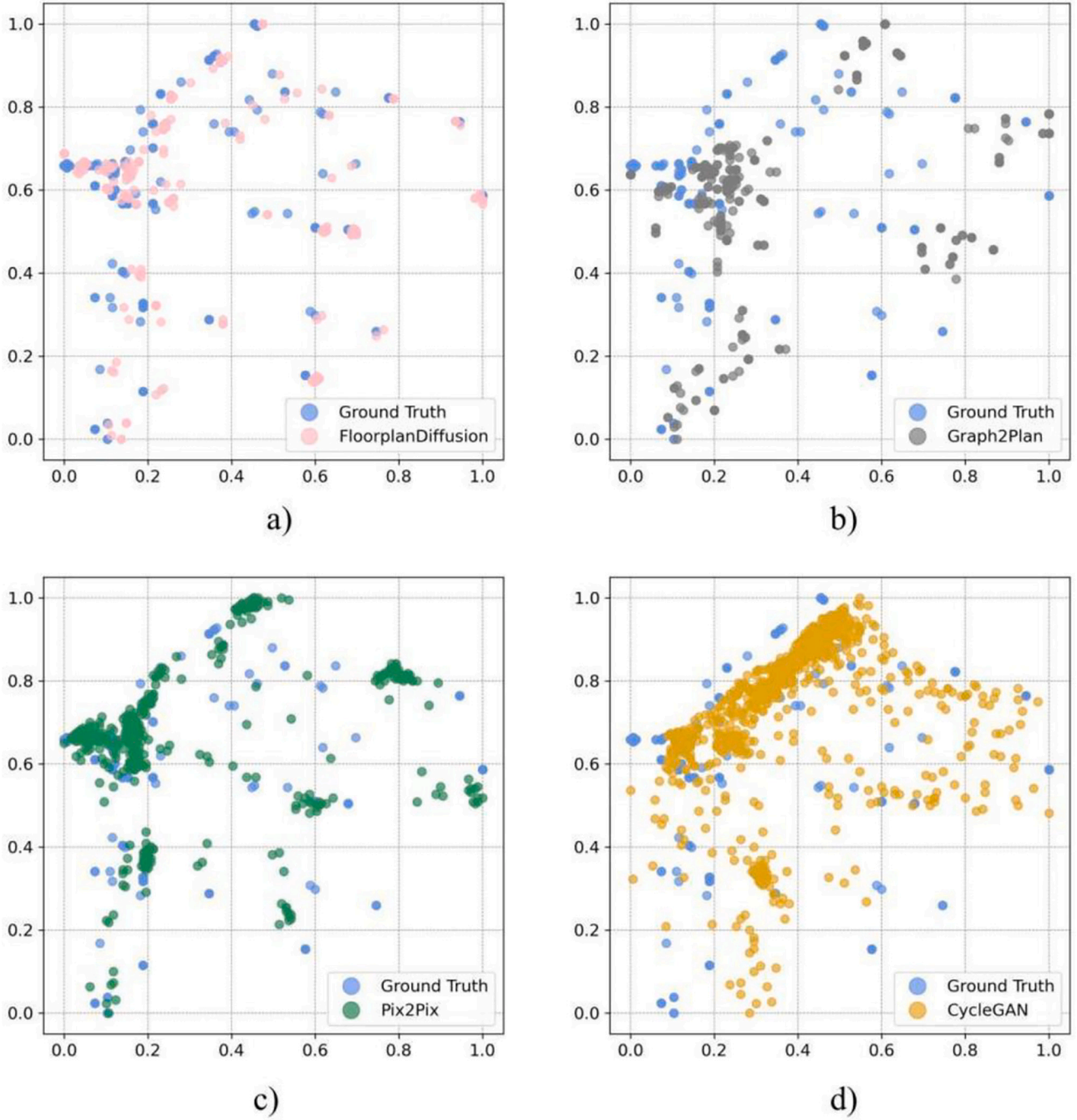


Fig. 11. PCA visualization results for generating FP features.

Table 2

Turing test results for the proposed FloorplanDiffusion model vs. real images.

	Model generation	Real images	All
Precision (average)	54.4%	45.6%	100%
Truth	50%	50%	100%
Increase ratio	+4.4%	-4.4%	0%

3.4.1.3. SSIM metric. SSIM is based on the assumption that the human eye extracts structural information from images and evaluates them by calculating the similarities in luminance, contrast, and structure between the original and reconstructed images. We require the generated images SSIM value to be as close to 1 as possible, indicating a high similarity with the real images in multiple aspects. The calculation

formula for the SSIM is as follows:

$$SSIM(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma$$

The first term $l(x, y)$ represents the calculated brightness similarity. The second term $c(x, y)$ represents the contrast similarity. The third term, $s(x, y)$, represents the structural similarity. In addition, $\alpha, \beta, \gamma > 0$ is used to adjust the weights of the aforementioned three components; however, it is typically set to 1.

3.4.2. Expert review

Many evaluation metrics have been proposed to evaluate the effectiveness of models comprehensively and accurately. However, these metrics cannot perfectly replace human subjective evaluations. Therefore, this study adopted the expert evaluation method used in the Turing

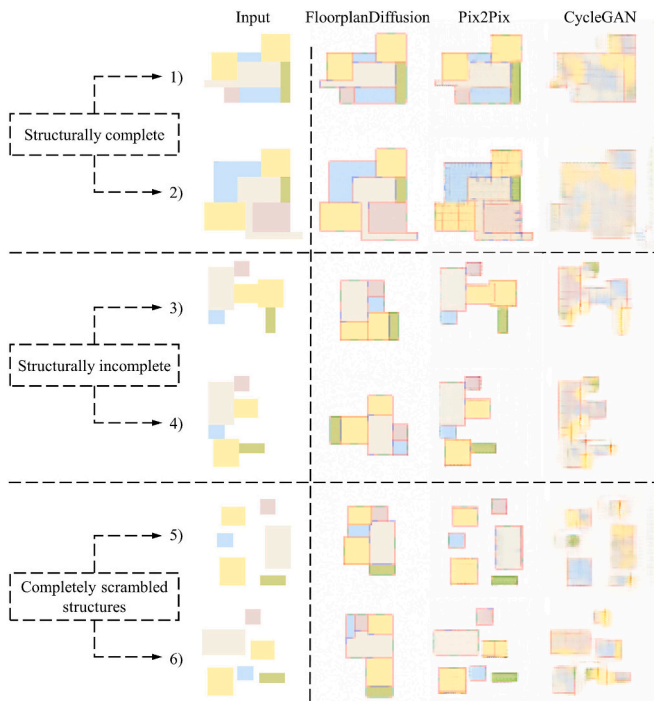


Fig. 12. Experts tested the results of the models' RFPs under different input conditions.

test [43] to comprehensively evaluate the effectiveness of the model. This involved experts who identify real FPs among a mixed collection of real plans and plans generated by our model. Specifically, 20 experts were invited to participate in the expert review. The experts included registered architects and graduate architecture students, who were first requested to provide five sets of input so that our model could generate floor plans, where each set of inputs had to be in the form of a bounding box to avoid the bias caused by human inputs. The dataset consisted of 200 FPs comprising 100 real and 100 generated samples. For each experiment, we randomly selected ten samples, which included five real and five generated samples. We subsequently combined the generated and real images and presented them in a shuffled order to the experts. The experts had to identify the five real images in each set. Each expert performed this task for five sets, i.e., in total reviewing 50 plans and selecting 25. (We remove the plans developed by the experts themselves for each expert during the assessment phase to avoid the experts recognizing that the plans were developed by themselves). Since our task involved choosing five samples from a pool of ten, the randomized correctness rate was 0.5, i.e., meaning that the model generated results that confused the experts. Lower precision and recall values indicate greater difficulty in distinguishing between the generated and real images. Hence, we utilized precision evaluations provided by multiple experts to assess the effectiveness of the generated images. Based on this, the strengths and weaknesses of the generated models were evaluated more comprehensively. This subjective evaluation method allows the generation results to be intuitively measured from human perspectives and provides a powerful means to assess model performance.

4. Results

4.1. Experimental results

Dataset and baseline model selection is crucial when evaluating a model. We used an RFP dataset generated using the RPLAN toolkit [44], which is the most commonly used dataset for residential room layout tasks. To improve training efficiency, the image resolutions were downsampled from 256 pixels in the original dataset to 64 using pixel

region relationships. The dataset was subsequently divided into training, validation, and testing sets, each with 70,126, 500, and 500 FPs, respectively.

We evaluated three different baseline models: Graph2Plan [19], Pix2Pix [20], and CycleGAN [21]. These baseline models were adopted in most previous studies on room layout or image generation tasks. The vast majority of the hyperparameters of these models were set according to the original authors' configurations, which performed well on the original dataset in their respective studies. After comprehensively testing the three baseline models, the results were used as important references for further analyzing performance.

4.2. Model training results

We compared and validated the proposed FloorplanDiffusion model with the baseline models. To eliminate evaluation bias arising from using a single metric and to provide a more comprehensive comparison, we used the FID, PSNR, and SSIM metrics for evaluation. A smaller FID value indicates a closer similarity between the generated and real images regarding feature distribution. A larger PSNR value indicates a higher reconstruction quality. A SSIM value closer to 1 indicates higher structural similarities between the generated and real images.

4.2.1. Evaluation metrics

The results of the compared evaluation metrics are listed in Table 1. The proposed model outperformed the other models regarding the FID and PSNR. Compared to the baseline models, the proposed model reduced the FID by 74.2% (Graph2Plan), 88.3% (Pix2Pix), and 93.8% (CycleGAN) and improved the PSNR by 7.7% (Graph2Plan), 7.9% (Pix2Pix), and 17.7% (CycleGAN). This indicates that the FloorplanDiffusion model outperformed other models regarding reconstruction quality and feature detail. All models performed similarly well regarding the SSIM metric, as indicated by the values close to 1. This is because the generated images have fewer color features, with only eight semantic blocks being colored. Therefore, the brightness and contrast of the generated images were similar, resulting in SSIM values close to 1. However, the subtle differences in the SSIM values reflect the structural similarity, indicating that the proposed model also outperformed the baseline models regarding structural similarity. In addition, to evaluate the stability of the generated models, we conducted a statistical analysis of the generated RFPs, as shown in Fig. 8. For the FID metric, we analyzed the FID values obtained through cross-validation. The PSNR and SSIM metrics were calculated separately by comparing the generated images with the ground truth. As shown in Fig. 9, the proposed model exhibited minimal fluctuations in the generated images, indicating high stability.

4.2.2. Model visualization

To validate the aforementioned evaluation metrics, we visually displayed the RFPs generated by our models (Fig. 8). The comparisons between the results of different models are shown in Fig. 10. The four rows represent different residential structures, where the left-most column of each row shows real images from the test set of the RPLAN dataset, and the next four columns contain the images generated by the proposed model and the baseline models. The results show that the FloorplanDiffusion model accurately reproduced the semantic information and color block layout of the real images, similar to the Graph2Plan model even without providing additional structural information. Although the Graph2Plan model can generate images with better structures based on graph structural information, it is somewhat blurry at the color block's edges, indicating inferior feature-detail performance. Furthermore, the images generated by the gPix2Pix and CycleGAN models may become unstable and collapse because of the GAN models' inherent limitations, where the interactions between the generator and discriminator lead to the models generating images that conform to human perception while not extracting or understanding the

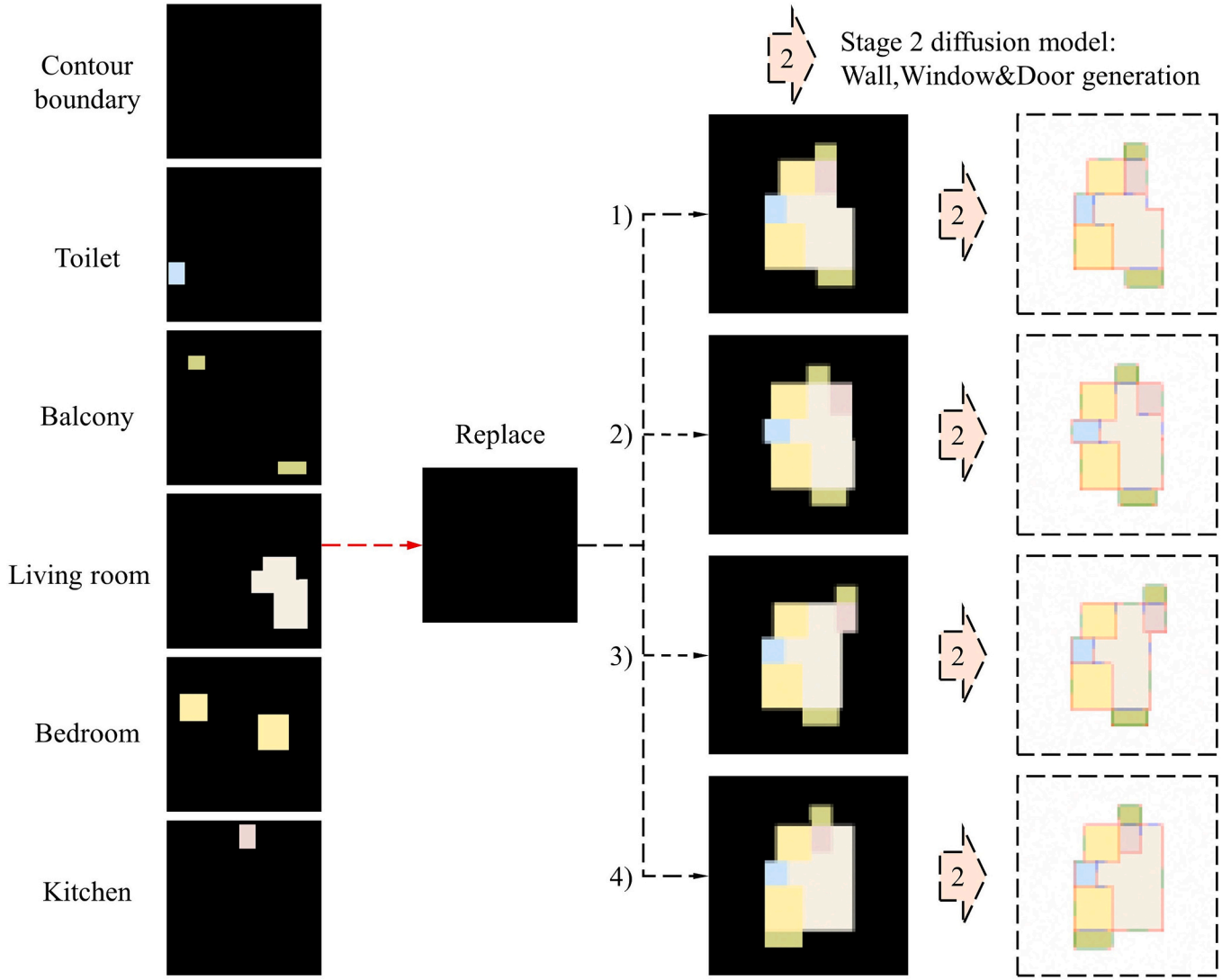


Fig. 13. Results of FloorplanDiffusion diversified model's reshaping despite the absence of semantic information.

semantic features. By adopting the DDPM diffusion structure, the FloorplanDiffusion model mitigates these issues and generates a more stable high-quality image.

4.2.3. Principal component analysis (PCA)

To explain the proposed model's superior performance, we utilized PCA to compare and analyze the features of the images generated by the FloorplanDiffusion model against those generated by the baseline models. For this, we first transformed the images into vector representations and subsequently used PCA to reduce the dimensions to extract the major components that significantly influenced the image features. By comparing these principal components, the similarity and quality of the images could be assessed. Fig. 11 shows the comparison results in which the FloorplanDiffusion model in the PCA space is closer to the real images, indicating that the proposed model captures the real image features more accurately. Compared to the other three models, the FloorplanDiffusion model evinced higher consistency and accuracy, resulting in more realistic and natural images with fine-grained details and more semantic information.

4.3. Expert Turing test results

The previous section compared the model's generation performance

on a test set extracted from the RPLAN dataset. Because of the similarity of many RFPs in the RPLAN dataset, a partial feature leakage may exist, i.e., the model may memorize some of the residential structural features of the RPLAN dataset during the training phase. To verify the proposed model's generalizability to other residential structure images, experts manually drew 100 input sets and we had our model generate complete RFPs (part of the outcomes are shown in Fig. 8). Then the experts were required to identify real plans from a mixed collection of real and generated plans (as described in Section 3.4) the results are shown in Table 2. The proposed model's precision is 54.4%, which is higher than the 45.6% for the real images. Therefore, the FPs generated by the FloorplanDiffusion model are highly accurate and resemble real residential plans, clearly proving the authenticity of the generated images.

To better understand the differences in image generation between the different models, we then had other models generate FPs using the input drawn by the expert, and some of the results are shown in Fig. 12. The Graph2Plan model did not fulfill the input requirements for this task; therefore, it was excluded. To demonstrate the model's ability to generate results under different input conditions, the 1)-2) rows, 3)-4) rows, and 5)-6) rows represent the proposed model's creative generation of an almost-complete house FP when given structurally complete, structurally incomplete, and completely scrambled structures as inputs. Simultaneously, the model ensured highly stable high-quality images

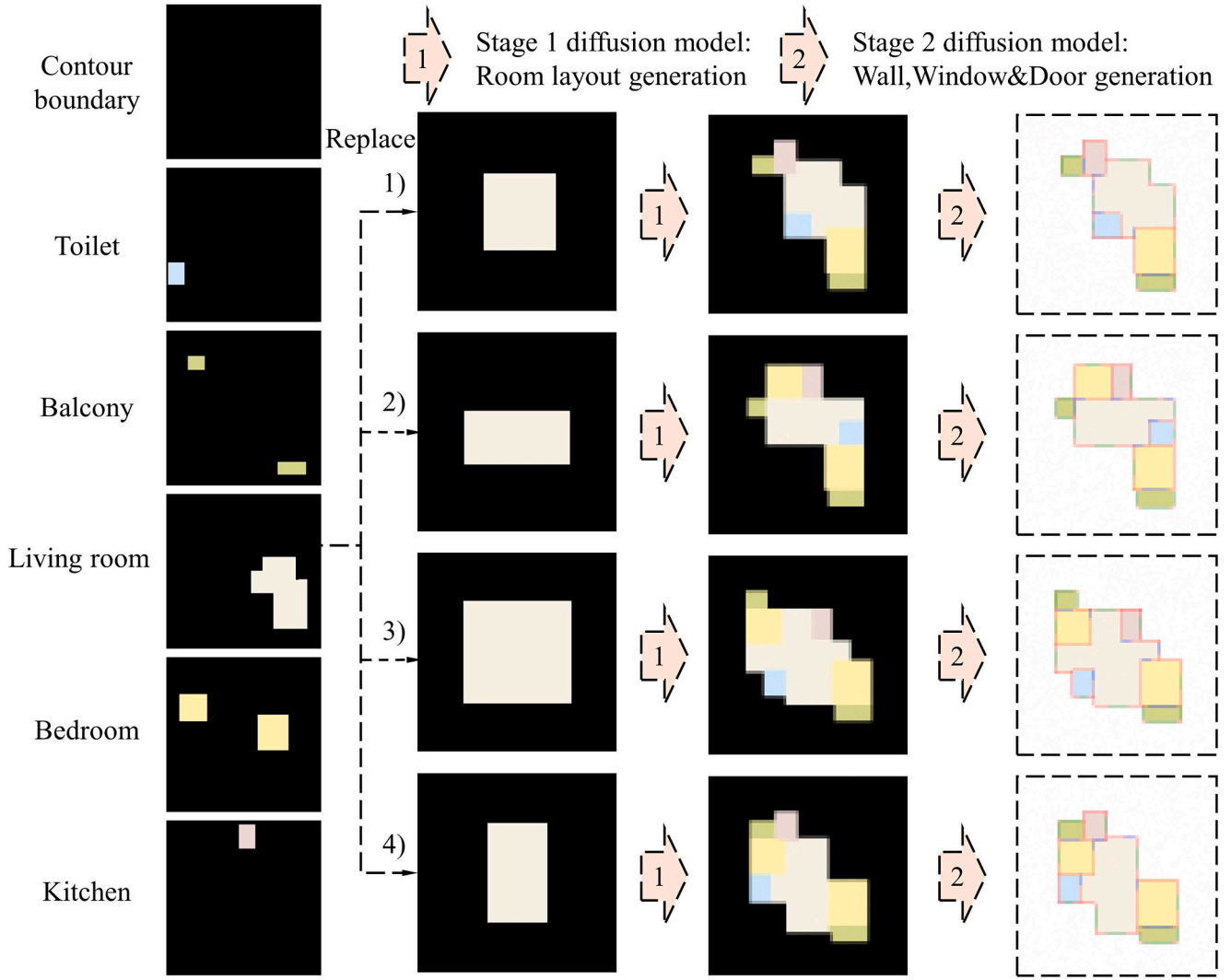


Fig. 14. Results of FloorplanDiffusion model control that transforms partial semantic segments into bounding box shapes. The user changes the shape of the living room into bounding boxes of different shapes for generation (in the second column, 1) is a small square, 2) is a long rectangle, 3) is a large square, 4) is a vertical rectangle). In the generated results, the living room conforms to the given bounding box shape.

when generating complete house FPs. This also reflects the high level of flexibility of the proposed FloorplanDiffusion model, which demonstrates excellent performance for various input forms.

5. Discussion

5.1. Model generation for multi-condition input

To ensure that models can adapt to a wider range of input data with the capability to generate more diversified outputs, we tested whether they could reshape semantic information diversely when some information is missing. We also tested whether, when given partial semantic information, the models could reshape it controllably.

First, how the models diversely reshape semantic information in the absence of this information is discussed. Specifically, we removed certain semantic information blocks (e.g., living rooms) that were essential to the FP. We examined whether the models could supplement these essential semantic information blocks using the generated results. Because the blocks were removed, there were many possibilities for filling them. Considering the missing living room information block as an example, the experimental results are shown in Fig. 13. The proposed model generated relatively complete FPs, even without living room information, thereby maintaining the stability and quality of the

generated images. Regarding diversity, owing to the absence of living room information, there are countless possible FPs. From this, it can be inferred that FloorplanDiffusion models can continue to adapt to various possibilities under these conditions.

Next, we examined the controllable reshaping of semantic information using models with partial semantic information. Specifically, we observed whether the models could generate controlled outputs for these semantic information blocks based on the bounding boxes of irregular semantic information blocks (e.g., living room). This condition drastically reduces the countless possible FPs to a limited number that comply with the given conditions. We experimented using the provided living room boundary box as a condition, and the experimental results are shown in Fig. 14. The FloorplanDiffusion models can generate complete FPs with similar shapes based on the given bounding box while maintaining the stability and quality of the generated images. Regarding diversity, the types of FPs generated are limited owing to the supplied conditions, however, the semantic information block of the living room still corresponds to the boundary conditions.

5.2. Model generation controllability experiment

Currently, two main types of models controllably generate FPs. The first is an end-to-end model based on a GAN, which cannot modify

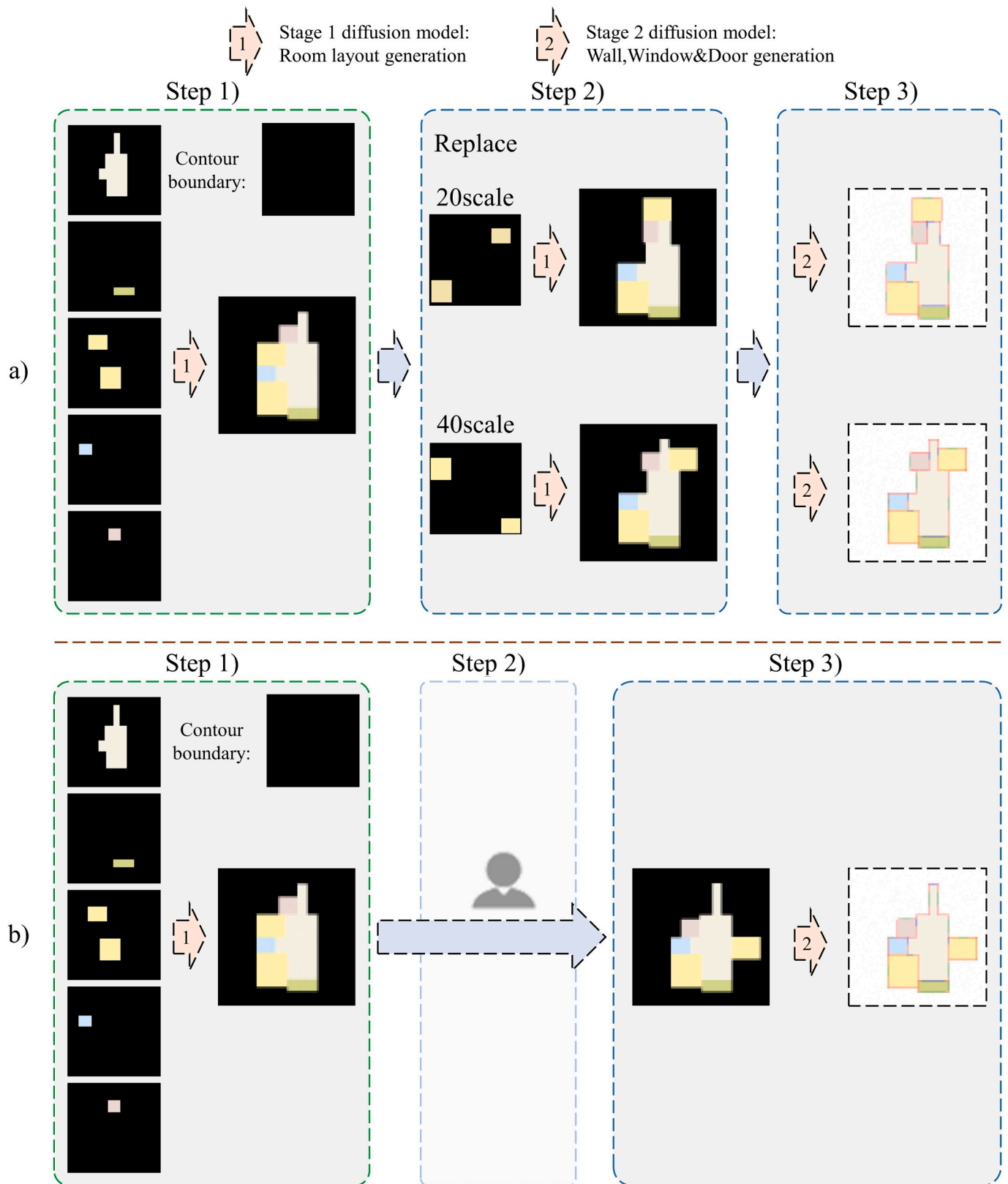


Fig. 15. Controllability experiments for the proposed model at different generation stages. In (a), users can generate FPs that meet more specific requirements by controlling the semantic blocks' approximate positions. (b) Users can generate their required floor plans by manually modifying the semantic block positions (Step 2) means that the user can manually modify the layout diagram generated in the first stage, including but not limited to the position and shape of the blocks and other features).

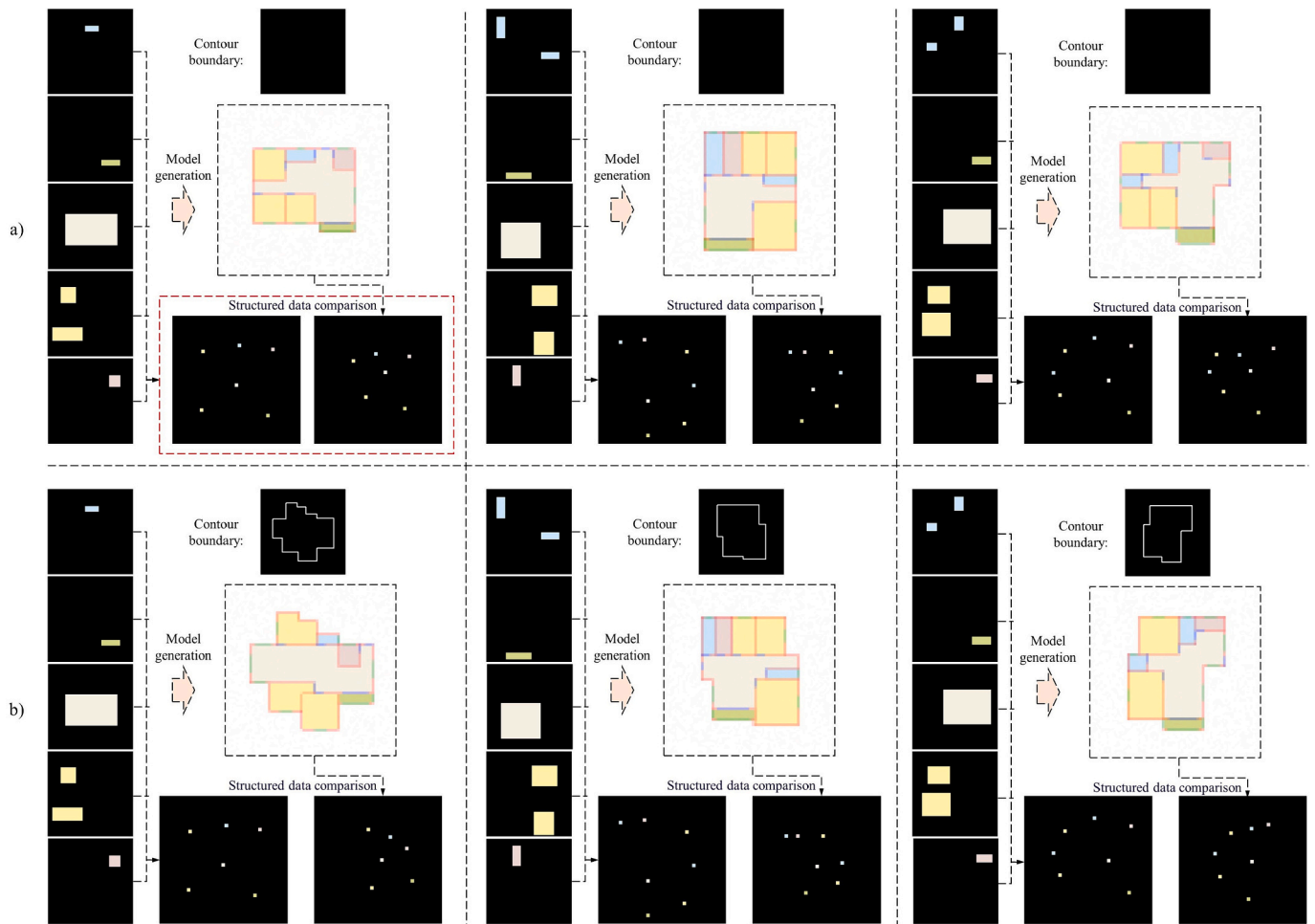


Fig. 16. Structural and shape information comparison. The first and second rows in the figure respectively are with/without contour edge information constraints. The left-hand column of each grid represents the input information given by the user (including certain structural data information and shape information). The picture on the left is the FP of the residential structure generated by our model. The two pictures below represent the input structural data information (left) and model-generated structural data information (right).

generated images. The other model is a GNN, which generates designs on a graph structure and can control the connection relationship of each room block, but editing others presents a significant difficulty.

To address this issue, we propose the FloorplanDiffusion model, which divides the end-to-end model generation into two stages. The FP generated in the first stage generates images that satisfy the requirements of a residential structure through flexible input. In the second stage, a complete RFP (walls, windows, and doors) is generated based on the image derived from the first FP. The flow is detailed in Fig. 15.

If users discover that the generated results of the model do not meet their requirements they can exercise control by doing the following. 1) Accurately position a semantic block: the block can be approximately aligned and the confidence of the position information can be set by adjusting the standard deviation of the Gaussian position information. The larger the standard deviation, the larger the range of the semantic block search. Using this method, a FP that meets user requirements can be regenerated using the first stage generated by the model. 2) Manually modify the block position by accurately moving it to the appropriate position. Subsequently, the modified FP that more closely meets the user's requirements is input into the second stage of model generation to create the complete RFP.

5.3. Model generation structure data comparison

We experimented to verify that the model operates under the conditions given by the user, i.e., whether the generated residential plan meets the structural data and shape information provided. We also conducted a series of experiments to verify that the model has certain constraint capabilities, i.e., under the conditions of a given contour edge, the contour edge of the generated residential plane can be strictly constrained. These experimental results are shown in Fig. 16.

From the figure, we can see that our model very well captures the shape information provided by the user. The structural data provided by the user is compared with the structural data generated by the model. The left-hand picture is the center position of each block where we extracted the picture given by the user. This position represents the user's preferential placement of each block and is similar to a bubble chart; the picture on the right shows the center position of each block in the results generated by our extracted model. The comparison results show that the RFP generated by our model can well restore the structural data information given by the user and strictly constrain the edges.

In summary, as hypothesized, our model generates a residential plan that conforms to the given structural data with or without contour edge constraints.

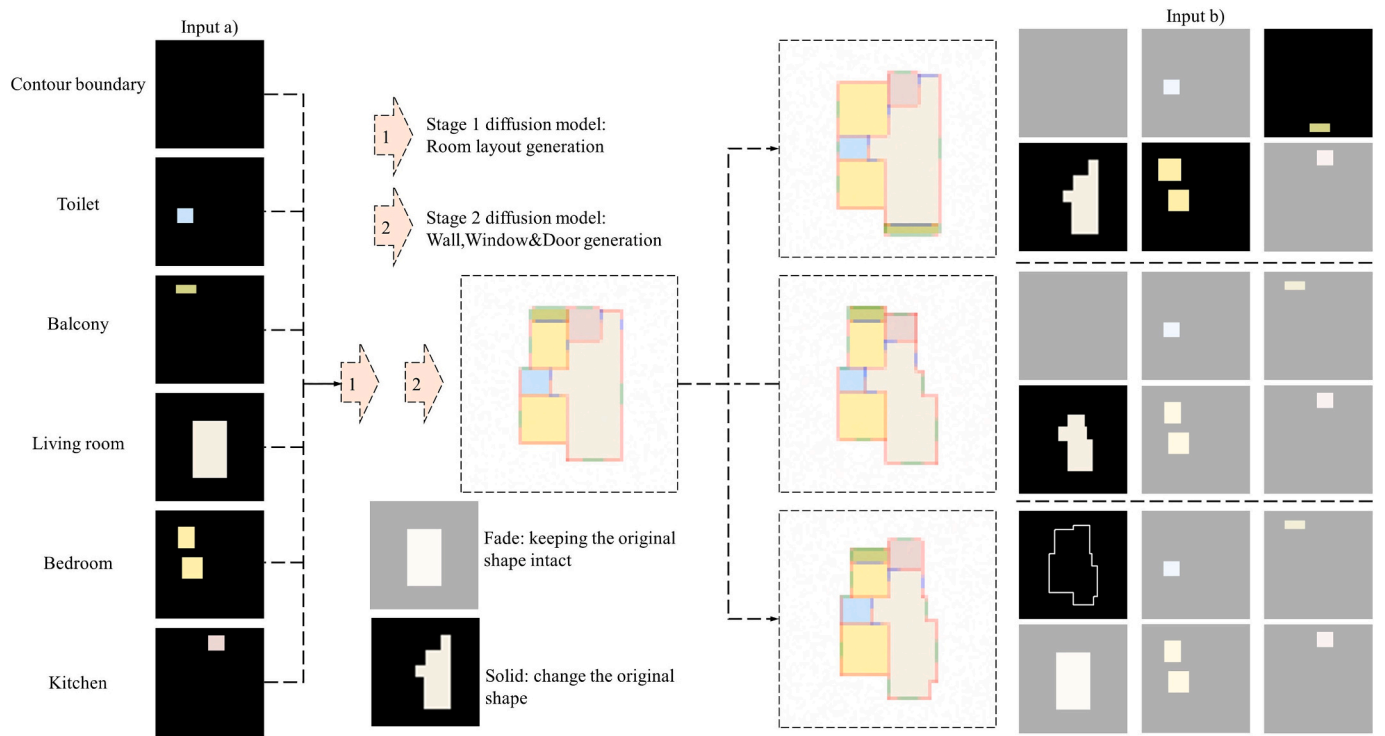


Fig. 17. Schematic diagram of the workflow. The left-hand side of the figure shows the initial input of the workflow a), the RFP is generated through the two stages described in the text, and then the generated RFP is modified by our modification input, which is shown in b) (the faded image in the figure indicates that the original shape remains unchanged, and the image that does not become a solid shape indicates that the original shape has changed).

5.4. Examples of inputs, outputs, and workflows

Inputs and outputs: our model operates by implementing two main generation stages: 1) room layout generation and 2) wall, window, and door generation. The room layout generation stage requires five inputs. In the first stage, the input of the room layout generation stage is one of the three types of images, i.e., the original value, the bounding box value, and the zero-value image, for five room blocks and one contour edge. Since we have used the conditional dropout mechanism the model recognizes the type of input and outputs a reasonable room layout graph. After obtaining the room layout from the first stage, we feed it into 2) the wall, window, and door generation stage. Thereafter, a complete RFP is output.

Workflow Example: For greater clarity, consider an example of designing from a layman's point of view, as shown in Fig. 17. At the beginning of the design process the user is uninspired and often provides little information, e.g., only the approximate size and location of the room blocks. Therefore, the input information is given as shown in Fig. 17 input a), and through the first and second phases of our model, a more reasonable RFP can be generated. Then the user can controllably edit each detail. Here we offer several examples of the process: 1) the shape of the living room is retained, and the upper left bedroom is enlarged while the balcony is moved to the bottom; 2) the shape of the living room is changed to the shape set by the user; and 3) the overall contour edges are changed to the shape set by the user. The input information for this step is shown in Fig. 17 input b), and the new RFP that conforms to the user's settings can be regenerated.

5.5. Experimental results

The following results were observed: 1) By performing semantic separation on the RFP images, the input conditions became more flexible. For example, Section 5.1, confirms that users can modify different semantic information blocks (such as complete deletions or conversions

to bounding boxes), thereby generating RFPs that satisfy their requirements. 2) By embedding the structured positional information into different semantic blocks and separating the image generation phase, the model's controllability was effectively improved. Section 5.2 confirmed that users can adjust the results roughly or precisely by modifying the input blocks in any generation phase to satisfy their specific requirements. 3) The DDPM structure is adopted to gradually generate stable RFPs based on feature distributions. Compared with the baseline models, Sections 4.2 and 4.3 describe that the proposed model performed better regarding stability, feature detail, and structural similarity in generating RFPs. 4) By removing semantic information to reduce the condition constraints, diverse RFPs can be generated. In Section 5.1, when completely removing or reducing the condition constraints, the model generated more diverse RFPs while maintaining the generated RFPs under the given semantic information conditions. 5) Our model can capture the structured positional features of the input. In Section 5.3, we visualized the structured positional features of the inputs and outputs, and the results demonstrate that the image inputs we employ can also contain graph-like structured positional features. 6) Our model can be a useful aid to non-professionals in designing an RFP. In Section 5.4, we described workflows that our model implements from a sketch design to output precisely controlled modifications.

While our model emphasizes user flexibility and controllability, there are certain input requirements the users must follow to reduce the chance of generating low-quality or infeasible RFPs. For example, there should be a minimum of one input condition (such as a living room, bedroom, or contour edge). There should be no curved elements (such as semi-circular or twisted features) due to the absence of curved residential features in the training dataset. Admittedly, more work needs to be done to further ensure the feasibility of each generated RFP. The autocorrection ability of our current model is limited if the user's input is unreasonable or violates the building codes. For instance, if the user draws a toilet larger than the bedroom, the model may still follow the direction and produce infeasible results. To cope with this, one possible

solution would be to develop an error-detection algorithm that can automatically identify potential problems in the generated RFP (i.e., a bedroom smaller than a toilet, or other building code violations) and suggest the user make modifications. Then the user can simply change the shape of the input images so that a new and more feasible RFP can be generated by our model. More detailed explorations regarding this will be done in the near future.

6. Conclusions and future work

Through extensive experimentation, we demonstrated that the proposed FloorplanDiffusion model can generate rational RFPs with multi-conditional image inputs. This confirms that our model lowers the skill threshold of residential floor plan design and allows non-professionals to design freely with our model. In addition, we demonstrated that our model generated controllable and diverse RFPs for users to select. It also allows for controlled, repeated editing of the floor plan until the client's needs are met. The above study confirms that the model achieved our initial research goals, i.e., stability, variety, controllability, and flexibility in generating RFPs. Finally, extensive evaluation and analysis of the proposed model clearly demonstrated its superior performance.

Currently, however, as with any contribution, the FloorplanDiffusion model has some limitations which should be acknowledged. For example, the embedding semantic information block position features remains discrete. This requires users to pre-constrain the semantic information block positions, i.e., in the data preprocessing phase, we must implement a well-defined structured location feature (e.g., a room block is searched within a 20 or 40 variance range around it) where the search range is a fixed discrete value, not a continuous value. Also, owing to the RPLAN dataset restrictions, the current FloorplanDiffusion model could only generate RFPs with certain geometric features, i.e., the generated rooms only contained features with 90-degree corners, but none such as semicircles, circles, or trapezoids. Our future work will therefore extend the proposed model to more general cases, such as generating RFPs with more diverse shapes and will consider the continuity of positional information features for semantic information blocks. With further advancements in generative learning, the proposed model will no longer be limited to generating two-dimensional RFPs but will be capable of generating three-dimensional RFPs. Furthermore, to provide new perspectives for future research directions on human-computer design collaboration, interpretable feature learning for RFPs must be further investigated.

CRedit authorship contribution statement

Pengyu Zeng: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Wen Gao:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Formal analysis, Data curation. **Jun Yin:** Writing – review & editing, Visualization, Investigation, Data curation. **Pengjian Xu:** Methodology, Investigation, Data curation. **Shuai Lu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

Shuai Lu reports financial support was provided by National Science Foundation of China. Wen Gao reports financial support was provided by National Science Foundation of China. Wen Gao reports financial support was provided by China Postdoctoral Science Foundation.

Data availability

Data will be made available on request.

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